This completes the proof of the theorem. □

Now, the algebra A is semisimple by Maschke's theorem, so the double centralizer theorem applies, and we get the following result, which goes under the name "Schur-Weyl duality" (as it was discovered by Schur and popularized by Weyl in his books *The theory of groups and quantum mechanics* and *Classical groups*; see Section 5.20).

Theorem 5.18.4. *(i) The image A of $k[S_n]$ and the image B of $\mathcal{U}(\mathfrak{gl}(V))$ in $\operatorname{End}(V^{\otimes n})$ are centralizers of each other.*

(ii) Both A and B are semisimple. In particular, $V^{\otimes n}$ is a semisimple $\mathfrak{gl}(V)$-module.

(iii) We have a decomposition of $(A \otimes B)$-modules

$$V^{\otimes n} = \bigoplus_{\lambda} V_\lambda \otimes L_\lambda,$$

where the summation is taken over partitions of n, V_λ are Specht modules for S_n, and L_λ are some distinct irreducible representations of $\mathfrak{gl}(V)$ or zero.

5.19. Schur-Weyl duality for $GL(V)$

The Schur-Weyl duality for the Lie algebra $\mathfrak{gl}(V)$ implies a similar statement for the group $GL(V)$.

Proposition 5.19.1. *The image of $GL(V)$ in $\operatorname{End}(V^{\otimes n})$ spans B.*

Proof. Recall that B is spanned by the elements $g^{\otimes n}$, $g \in \operatorname{End} V$. Denote the span of $g^{\otimes n}$, $g \in GL(V)$, by B'. Let $b \in \operatorname{End} V$ be any element.

We claim that B' contains $b^{\otimes n}$. Indeed, for all values of t but finitely many, $t \cdot \mathrm{Id} + b$ is invertible, so $(t \cdot \mathrm{Id} + b)^{\otimes n}$ belongs to B'. This implies that this is true for all t, in particular $t = 0$, since $(t \cdot \mathrm{Id} + b)^{\otimes n}$ is a polynomial of t. More precisely, if f is a linear function on $\operatorname{End}(V^{\otimes n})$ that vanishes on B' then $f((t \cdot \mathrm{Id} + b)^{\otimes n})$ is a scalar-valued polynomial of t which vanishes for almost all $t \in k$, hence is identically zero.

The rest follows from Lemma 5.18.3. □